Author-review draft; numerical result and author declarations pending. Not submitted.

7 November 2026

Dear Editors,

Please consider our manuscript, “Lunar tidal potential measured on a coastal–inland optical-clock baseline”, for publication as an Article in *Nature*.

Optical clocks offer a way to observe gravitational potential through relative ticking rates. For time-varying measurements, a central difficulty is attribution: astronomical tides and laboratory conditions can both produce variations near the solar day and its harmonics. Our manuscript addresses this problem in a comparison of strontium and ytterbium clocks in Shanghai and Wuhan, connected by a 1,350-km fibre link across a 692-km coastal–inland baseline.

The study focuses on the lunar semidiurnal tide, whose phase moves through local time. We combine several campaigns in a joint astronomical model and quantify how specified daily disturbances project onto the lunar response. The principal measurement is [insert recalculated lunar response, uncertainty definition and confidence interval]. Local and conditional spectra, single-component subtraction and sampling-window injections test whether neighbouring spectral structure can arise from leakage. Environmental records, transmission monitoring and dependence-aware uncertainty checks further delimit the interpretation. We distinguish common from campaign-dependent disturbance phases and report ocean-loading model sensitivity separately from statistical uncertainty.

We believe the work will interest readers in precision measurement, gravitation and geophysics because it addresses how a quantum sensor network can identify a weak geophysical signal in the presence of competing periodic variation. Its contribution is a measured lunar response with a quantitative assessment of attribution, rather than a general claim that all daily laboratory effects have been excluded.

The summer observations overlap the earlier absolute frequency-ratio study by Hou and colleagues (arXiv:2610.34250). The manuscript states this relationship and distinguishes the extended record and grouped tidal analysis from that study; shared-data estimates are not combined as independent evidence. We also discuss the related European-network observations reported by Stahl and colleagues (arXiv:2610.39635).

[Before submission, insert author-confirmed statements on approval by all authors, exclusive consideration, related manuscripts, any prior editorial discussions, competing interests and data/code access.]

Thank you for considering our work.

Yours sincerely,

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[Institution and email]